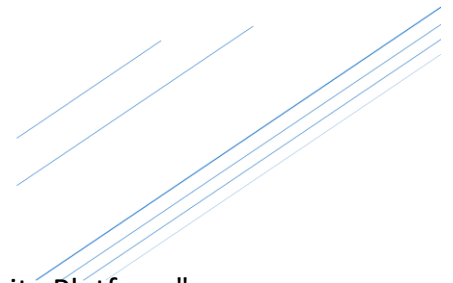


RAJARATA UNIVERSITY OF SRI LANKA

Department of Information and Communication Technology

ICT 1209: WEB TECHNOLOGIES MINI PROJECT — FINAL REPORT



"Yummy.lk — Interactive Digital Recipe Book & Community Platform"

Theme: Digital Recipe Book

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Date : 2026/09/23

GitHub Repository: github.com/DushanThiwanka/digital-recipe-book

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1. Project Overview

1.1 Purpose

Yummy.lk is an interactive, database-driven web application designed to deliver a simplified yet powerful platform for culinary learning and community engagement. The system enables users to discover, search, and filter recipes dynamically, while also allowing registered members to contribute their own recipes and interact with a wider community of home cooks.

1.2 Target Audience

- Housewives — individuals seeking diverse, healthy, and innovative daily meal inspirations for their families.
- Beginners & Non-Cooks — university students or young professionals living alone who need step-by-step, easy-to-follow guidance to cook basic meals.
- Food Enthusiasts — community members who want to share unique recipes, receive feedback, and interact with other home cooks.

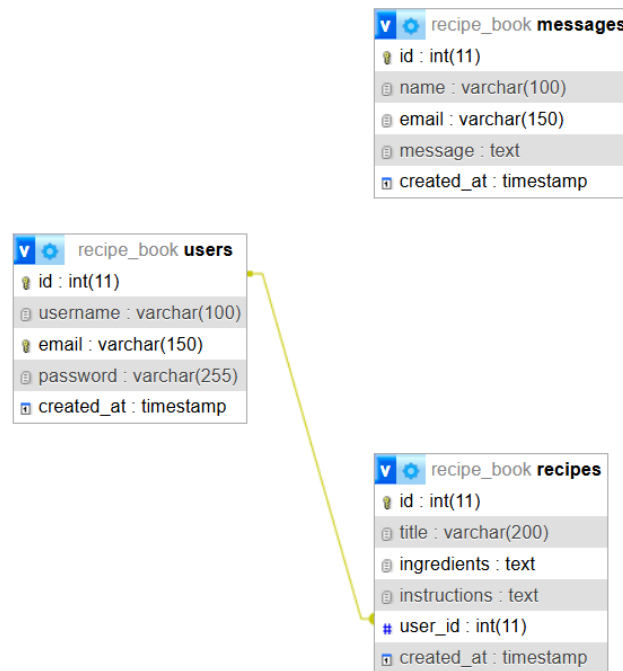
1.3 Key Features

- Dynamic Recipe Database & Search — a user-friendly search system that filters recipes instantly by category or ingredients using client-side JavaScript.
- Recipe Input & Management Engine — an interactive form allowing registered users to contribute their own recipes (title, ingredients, preparation steps) directly to the database.
- Interactive Community Feedback Portal — allows visitors to post comments and queries regarding specific recipes.
- Secure User Authentication — registration and login handled securely via backend PHP logic using BCrypt password hashing.

2. System Design

2.1 Database Schema

The application uses a MySQL database named `recipe_book`, consisting of three core tables: `users`, `recipes`, and `messages`. The `recipes` table maintains a foreign-key relationship with `users` through the `user_id` field, linking each submitted recipe to its author.



Entity-Relationship Diagram of the recipe_book database

Table	Field	Type	Description
users	id	int(11)	Primary key, auto-increment
users	username	varchar(100)	Unique display name
users	email	varchar(150)	User email address
users	password	varchar(255)	BCRYPT-hashed password
users	created_at	timestamp	Account creation time
recipes	id	int(11)	Primary key, auto-increment
recipes	title	varchar(200)	Recipe title
recipes	ingredients	text	List of ingredients
recipes	instructions	text	Preparation steps
recipes	user_id	int(11)	Foreign key → users.id
recipes	created_at	timestamp	Submission time
messages	id	int(11)	Primary key, auto-increment
messages	name	varchar(100)	Sender's name
messages	email	varchar(150)	Sender's email

Table	Field	Type	Description
messages	message	text	Message content
messages	created_at	timestamp	Submission time

2.2 Folder Structure

The project follows a modular folder structure that separates presentation, logic, and configuration concerns:

```

project/
├── css/
│   └── style.css           # Stylesheets and visual design
├── js/
│   └── script.js          # Client-side validation and interactivity
├── images/                # Static image assets and icons
├── includes/
│   ├── db.php             # PDO database connection handler
│   └── functions.php      # Reusable helper and security functions
├── auth/
│   ├── register.php       # User sign-up engine with BCrypt
│   ├── login.php          # Credential check & session initiation
│   └── logout.php         # Session destruction handler
├── contact.php            # Inquiry submission form
├── index.php              # Main landing page
├── dashboard.php          # Authenticated user control view
├── README.md              # Deployment and system setup guide
└── database.sql           # Exported database migration script

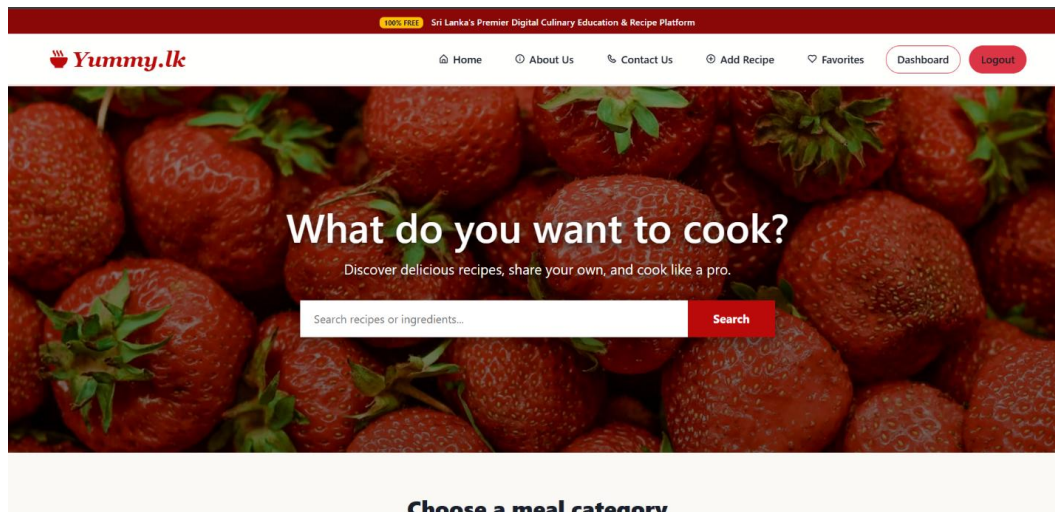
```

Project folder structure

3. Implementation

3.1 Frontend Structure & Styling

- Built with HTML5 semantic elements, styled with Bootstrap 5 and modular custom CSS (css/style.css).
- Incorporates responsive Flexbox and CSS Grid layouts, media-query breakpoints, and dynamic hover cards.



Yummy.lk homepage — hero search section and navigation bar

3.2 Client-Side Logic (js/script.js)

- Real-time query search matching recipe titles and ingredient lists without full-page reloads.
- Category selection, drink sub-category filters, and dynamic DOM rendering.
- Client-side validation enforcing input sanity alongside server-side checks.

3.3 Backend Logic & Security

Database Connection (includes/db.php)

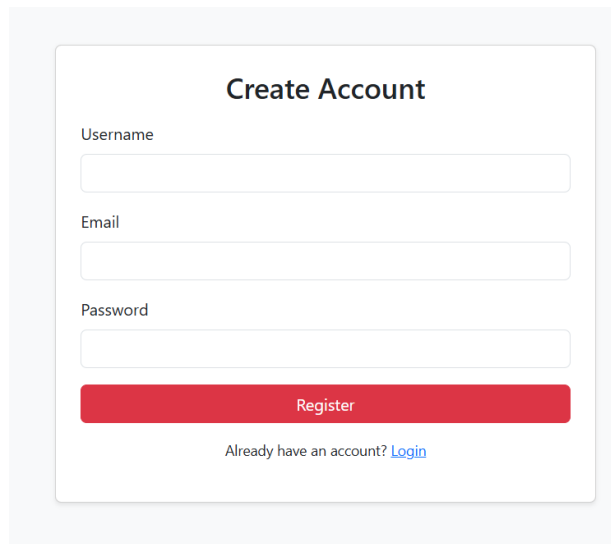
A centralised PDO (PHP Data Objects) implementation configured with `ERRMODE_EXCEPTION` and disabled emulation of prepared statements, eliminating SQL injection vulnerabilities.

Helper Modules (includes/functions.php)

Dedicated functions for input sanitisation (`htmlspecialchars()`) and session access validation (`isLoggedIn()`).

Registration Handling (auth/register.php)

Validates required parameters, performs duplicate credential checks, hashes passwords via `password_hash($password, PASSWORD_BCRYPT)`, and executes prepared SQL inserts.

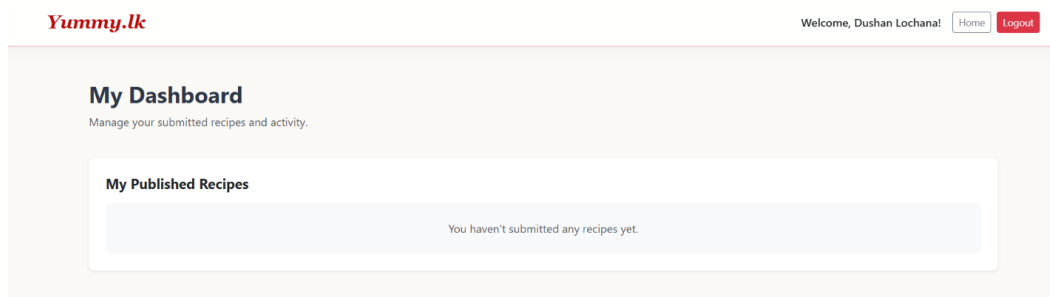


The image shows a 'Create Account' form. It has three input fields for 'Username', 'Email', and 'Password'. Below these fields is a red 'Register' button. At the bottom, there is a link that says 'Already have an account? [Login](#)'.

User registration page

Authentication Flow (auth/login.php)

Verifies submitted credentials against stored hash values via `password_verify()`. Invokes `session_regenerate_id()` upon success to prevent session-fixation attacks and redirects authorised sessions to `dashboard.php`.



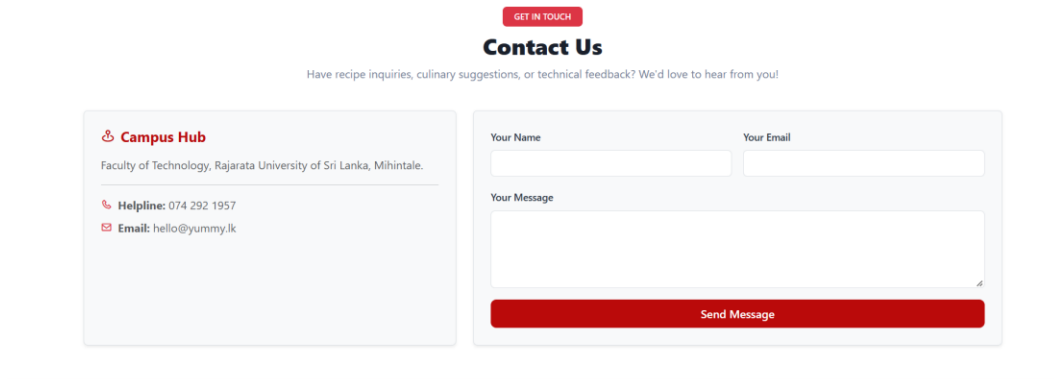
Authenticated user dashboard

Session Termination (auth/logout.php)

Flushes session superglobals, invalidates session cookies, destroys the server-side session, and redirects cleanly to `index.php`.

Contact Integration (contact.php)

Processes incoming user queries, stores them in the messages table via prepared statements, and returns confirmation feedback to the user.



The image shows a 'Contact Us' form. At the top, there is a red button labeled 'GET IN TOUCH'. Below it, the heading 'Contact Us' is displayed in bold. A subtext reads: 'Have recipe inquiries, culinary suggestions, or technical feedback? We'd love to hear from you!'. The form is divided into two main sections. The left section, titled 'Campus Hub' with a red icon, contains the text 'Faculty of Technology, Rajarata University of Sri Lanka, Mihintale.' and contact information: 'Helpline: 074 292 1957' and 'Email: hello@yummy.lk'. The right section contains input fields for 'Your Name' and 'Your Email', a large text area for 'Your Message', and a red 'Send Message' button at the bottom.

Contact Us form

4. Challenges and Solutions

4.1 Folder Restructuring & Broken File Inclusions

Challenge: Moving backend files into `auth/` and `includes/` initially caused stream inclusion errors ("Failed opening required `'../includes/functions.php'`").

Solution: Absolute path resolution using the PHP magic constant `__DIR__` (`require_once __DIR__ . '/../includes/functions.php'`) was implemented to eliminate directory-traversal ambiguities.

4.2 Session Persistence & Secure Redirects

Challenge: User sessions required proper state management across directory depths while enforcing security standards.

Solution: Integrated `session_regenerate_id(true)` upon successful verification, standardised `session_start()` invocations, and built automated session gates on `dashboard.php`.

4.3 Form Submissions & Database Field Mapping

Challenge: Ensuring both client-side JavaScript validation and traditional server-side POST submissions operate without conflict.

Solution: Standardised backend form processing directly on `contact.php` with prepared queries while maintaining responsive UX alerts.

5. Individual Contribution

Y. Y. G. D. T. Lochana — ITT/2024/063 (Frontend UI/UX Developer)

- Frontend Layout & Design: built the fully structured HTML pages (Home, Menu, Details) and styled them using CSS3 and Bootstrap 5.
- Client-Side Logic: implemented JavaScript features including dynamic content updates, event handling, form validation, and local-storage interactions.

A. M. I. C. Abeykoon — ITT/2024/003 (Backend & Database Developer)

- Database Setup: designed the MySQL database schema (recipe_book) and created the required tables (users, recipes, messages).
- PHP Integration: developed the user authentication system (login/register with BCrypt) and the contact-form backend logic using prepared statements.

6. Setup Instructions

Server Setup

- Download and install the XAMPP stack (PHP 8+, Apache, MySQL).
- Start the Apache and MySQL services via the XAMPP Control Panel.

Project Deployment

- Place the project directory inside the XAMPP root folder: C:\xampp\htdocs\yummylk\.

Database Migration

- Access phpMyAdmin at <http://localhost/phpmyadmin/>.
- Create or import the database using the supplied database.sql script located in the project root.
- Ensure database connection configuration in includes/db.php matches your local environment credentials (Host: localhost, User: root, Password: "", Database: recipe_book).

Execution

- Launch your web browser and navigate to: <http://localhost/yummylk/index.php>

7. References

- [1] PHP Documentation Group. (2026). PHP: PDO – Manual & Password Hashing Functions (password_hash, password_verify). [php.net](https://www.php.net).
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